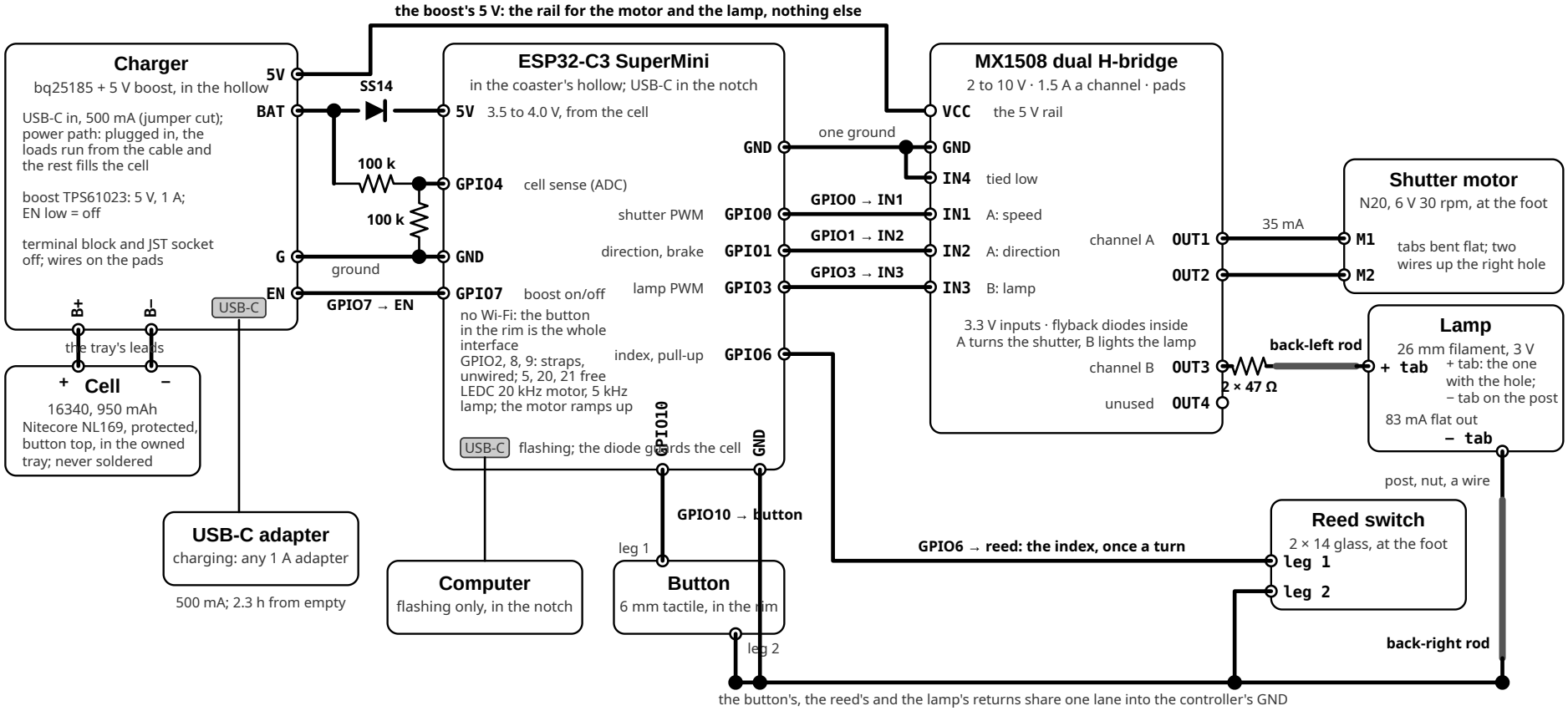
# Desk lighthouse: every connection on one sheet, the power chain first

A dot is a joint. A line that hops over another does not touch it. Thick = 30 AWG wire you solder. Thin = a lead or a cable. Brass-filled = a rod of the frame, carrying current. Left to right: cell, charger, controller, driver, loads. Zero crossings.




### The budget: from the cell, and on the boost's 5 V rail

| | |
|---|---|
| **cell** | 129 mA with the beacon turning and the lamp at 70 %: about 7 h · 112 mA lamp alone: 8 h · 20 mA asleep, boost off: 2 days |
| **rail** | 70 mA typical on the boost's 5 V, 119 at most in normal running; the boost stalls if more than 200 mA is switched on in a step, so the motor ramps |
| **motor** | 35 turning at 32 % duty (8 rpm); about 400 if the shutter jams, which the reed's silence reports |
| **lamp** | 83 flat out, 58 at the 70 % default; 249 mW in the lantern, 81 mW in each resistor; up one rod, down another |
| **controller** | 20 from the cell through the diode, no Wi-Fi; the divider 21 µA |
| **charge** | 500 mA with the jumper cut, about 2.3 h from empty; plugged in, the lighthouse runs from the cable and the cell fills with the rest |

Never: the cell without the diode between it and the controller · the charger's jumper uncut · the lamp without its resistors · the live rod touching any other brass (meter it) · an iron near the cell or the magnet.

Rev D, drawn 2026-11-09 evening for approval, not built. 0 crossing(s) on this sheet. Sources: the charger's pinouts page, the SuperMini notes (5V = VBUS), the MX1508 datasheet, the satellite's record of the same cell, charger and diode, the